12 Accessing the hardware

The only part we've not dealt in this book till now is how to access the iPhone's hardware. We've used the applications, camera, contact list etc already in our projects. Now in this chapter, we will show you how to use specialized classes in the iPhone SDK to access hardware devices, such as the accelerometer, GPS and how to obtain location information using GPS, cell towers and wireless hotspots.

Let's start with an application that uses the accelerometer for some purpose other than changing the rotation of the screen, which we've already shown you how to. The accelerometer of the iPhone allows the device to detect the orientation of the device and adapts the content to suit the new orientation. Even the camera relies on the accelerometer to tell it whether you are taking a picture in portrait or landscape mode. It measures the acceleration of the device relative to freefall in three different axes: X, Y,and Z. To make things clear, here's a table to show the values of X, Y and Z for various positions of the iPhone screen.

Position	X	Y	Z
Vertical upright position	0.0	-1.0	0.0
Landscape Left	1.0	0.0	0.0
Landscape Right	-1.0	0.0	0.0
Upside Down	0.0	1.0	0.0
Flat Up	0.0	0.0	-1.0
Flat Down	0.0	0.0	1.0

In general, the value of X will increase if the device is held upright and moved to the right quickly. Similarly, if moved to the left quickly, X will decrease. The same principle applies for the Y and Z directions too. The accelerometer used on the iPhone gives a maximum reading of about +/- 2.3G with a resolution of about 0.018 g.

Now that your concepts about the accelerometer are clear, lets get to work with some actual code. Make note that this code cannot be properly tested on the iPhone Simulator, so its recommended that you get a developer license and test it on an actual hardware. We will first begin with programmatically accessing the data returned by the accelerometer, which can then be used for motion detection algorithms in applications like games.